Cover Letter

To: Editor-in-Chief, ACM Transactions on Computing Education
From: Wenbin Hu, School of Computer Science, Xijing University
Date: July 30, 2026
Subject: Manuscript Submission — "Automated Programming Verdict Classification from Submission Metadata: Comparing Machine Learning and Large Language Models in Competitive Programming"

Dear Editor,

We are pleased to submit our manuscript, "Automated Programming Verdict Classification from Submission Metadata: Comparing Machine Learning and Large Language Models in Competitive Programming," for consideration in ACM Transactions on Computing Education.

Why This Paper Fits TOCE. Automated feedback is central to computing education, yet most existing tools require source code access. We address a deceptively simple question: **what verdict information can be recovered from metadata alone?** Our answer reveals a platform-encoding boundary — a natural limit that neither ML nor LLMs can overcome without source code. The paper provides educators with cost-effective verdict-screening tools (2 ms inference, $0.01 per 1,000 classifications) and an empirical framework for comparing supervised ML and zero-shot LLM approaches.

Novelty and Contributions.

- **First controlled ML–LLM comparison** for programming verdict classification from metadata alone, with identical protocols across three ML models (Random Forest, Gradient Boosting, Logistic Regression) and two LLM configurations (DeepSeek-V3, Qwen2.5:3B).
- **The platform-encoding boundary**: CE, TLE, and MLE (18.4% of submissions) are deterministically recoverable from execution metadata (F1 $\geq$ 0.89) because platform measurement thresholds define these outcomes. The WA$\leftrightarrow$RE boundary is not metadata-separable (RE F1 = 0.52), indicating where source-code analysis may be required.
- **Statistical and deployment rigor**: Holm-Bonferroni-corrected McNemar tests, ablation analysis, Gini importance, leave-one-user-out cross-validation (82.5% accuracy), and cross-difficulty generalization (71–94% across problem difficulties).
- **Open dataset and code**: `https://github.com/huwenbin123huwenbin/programming-error-clas`

Key Findings. Gradient Boosting (95.02%) outperforms DeepSeek-V3 zero-shot (76.65%) by 18.4 pp; even with identical features GB (92.0%) surpasses DeepSeek-V3 by 15.4 pp. Execution time is the dominant predictor (8.8 pp accuracy drop when removed). We note that this comparison is asymmetric: ML models were trained on 10,688 labeled samples while LLMs received no task-specific training.

Relevance to TOCE Scope. This work speaks to TOCE's interest in: (1) automated assessment tools for resource-constrained settings; (2) feedback automation characterizing which verdicts benefit from immediate feedback (CE, TLE, MLE) and which require human expert review; (3) empirical computing education research with transparent statistical analysis.

Submission Materials. Manuscript: 28 pages, 8 tables, 7 figures. Highlights: 5 key contributions. Supplementary: per-class F1 scores, Cohen's h effect sizes, balanced accuracy, confusion matrices, ROC/PR curves.

We confirm this manuscript is original, has not been published elsewhere, and is not under review at any other venue.

Best regards,

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